Fossils everywhere 14a

Types of fossils

1 Read the text about fossils and write the missing words into the spaces. The missing words are supplied below.

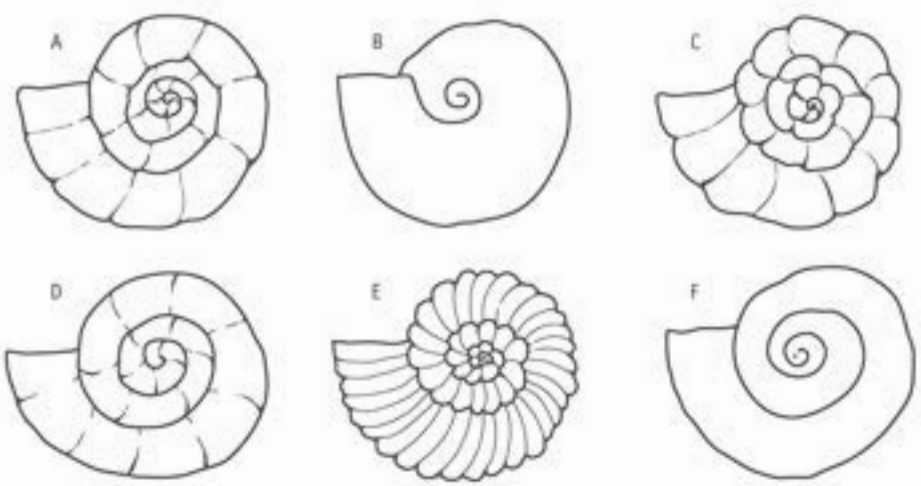
There are three types of fossils: Originals, moulds and casts. When an organism or some of its parts are covered by _____ like sand or mud, the sediments protect and preserve _____. The preserved organic part is an original fossil. Eventually the sediments harden into _____. Often the organic part _____ thus leaving a cavity in the rock material. This _____ is called a *mould;* detailed imprints of the organism can be seen in the rock. Sometimes this cavity is filled with materials which eventually harden into rock. This new material forms a cast of the original _____. Many things can become _____. Usually the hard parts of organism like *bones, teeth* and *tree trunks* become _____. But sometimes soft parts like hair, *feathers* and *leaves* also develop _____. *Animal droppings* may on occasions also become fossils, _____ *footprints*. It is often _____ for the untrained eye to recognise a fossil.

as well as	cavity	difficult	fossilised	fossils	into fossils	sedimentary rock
organism	rots away	sediments	the organic parts			

2 Using the italicised words from the text above, draw an example of each fossil.

Ordering fossils

3 Scientists believe that older fossils are simpler than younger ones. Look at the fossils of various ammonites. Ammonites were most common during the Mesozoic period, 248–65 million years ago. The animal that lived inside the shell was related to the present-day octopus and squid. Ammonites varied greatly in size, from those as small as a coin, to others as big as a tyre. Determine which is the oldest fossil (1) and which is the youngest fossil (6). Then determine the order of the other ammonites and indicate it by writing the numbers next to the drawings.



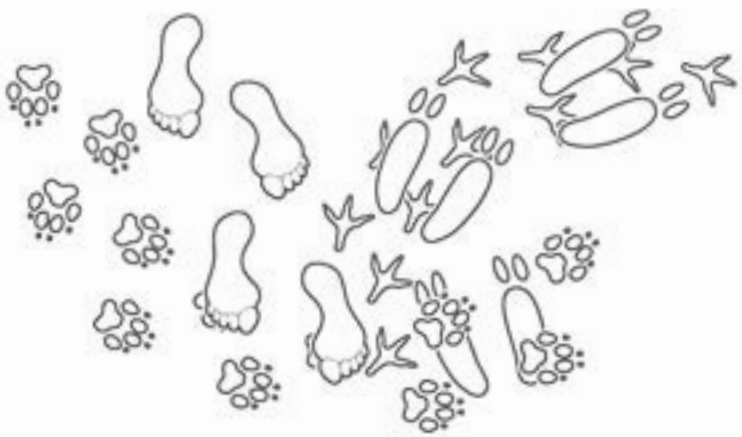
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Fossils everywhere

14b

Sequencing imprints

4 Look at the prints in the mud. The prints come from a human, a seagull, a dog and a kangaroo.



a Determine which organism walked on the mud first. Justify your answer.

_____

_____

b Write down the names of the visitors in order, starting with the first visitor.

_____

_____

c Imagine the sequence of events that may have happened at this muddy place.

_____

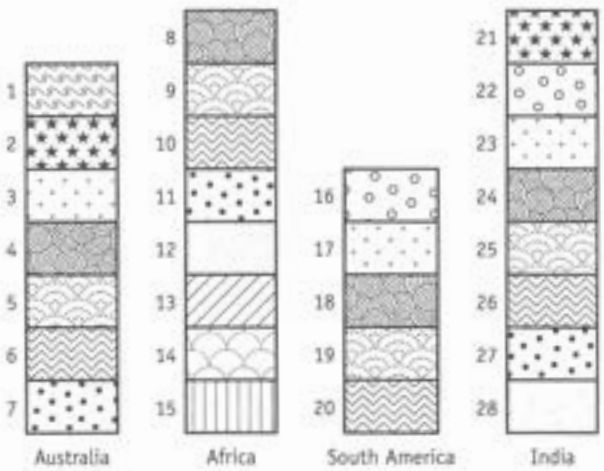
_____

_____

Matching rock layers

Scientists believe that rock layers beneath other rock layers are older than all the layers above. They also believe that rock layers which come from different places but contain the same kind of fossils are the same age.

5 Study the rock columns on the right and then perform the tasks below.



a Identify several index fossils between any two rock columns and join the matching rock layers with a line.

b Determine the oldest and youngest rock layer and label them.

c Deduce the order of layers from oldest to youngest and write their corresponding numbers down.

Oldest layer _____ youngest layer

d Suggest reasons why the highest column does not have the youngest rock layer at the top.

_____

_____

_____

e Explain why there is a rock layer missing in the column from Australia.

_____

_____

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